

Kyuewon Kim

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Ph.D. Student, Department of Information Display, Kyung Hee University
Human-Centered Extended Reality Lab (Advisor: Prof. Uijong Ju)

✉ kyuewonkim@khu.ac.kr  github.com/kyuewonkim  linkedin.com/in/kyuewonkim  linkedin.com/in/kyuewonkim

Summary

Human-Computer Interaction (HCI), Extended Reality (XR), Digital Therapeutics (DTx), Healthcare

Education

Kyung Hee University

Ph.D. / Department of Information Display

Mar. 2026 ~ Present

Kyung Hee University

M.S. / Department of Information Display

Mar. 2024 ~ Feb. 2026

Thesis: Research on Cybersickness Mitigation Using Peripheral Vision Display Technology

Korea University (Transfer)

B.S. / Department of Display and Semiconductor Physics

Mar. 2020 ~ Feb. 2024

Gyeongnam National University of Science and Technology

B.S. / Department of Mechanical Engineering (Night Program)

Mar. 2016 ~ Feb. 2020

Publications

[P2] **Kyuewon Kim, Uijong Ju**: Manipulating and Predicting Subjective Visual Fatigue While Watching Video Content: A Whole-Brain Functional Connectivity Analysis, *Displays*, 2026.

[P1] **Kyuewon Kim, Uijong Ju**: Manipulating and Predicting Subjective Visual Fatigue While Watching Video Content: A Whole-Brain Functional Connectivity Analysis, *IEEE Transactions on Visualization and Computer Graphics*, 2026.

Conference

Oral

[CO1] **Kyuewon Kim, Uijong Ju**: Fading the Periphery: Peripheral Brightness Rolloff and Resolution Reduction for Cybersickness Mitigation in Virtual Reality. *IEEE VR 2026*.

Poster

[CP1] **Kyuewon Kim, Uijong Ju**: Fading the Periphery: Peripheral Brightness Rolloff and Resolution Reduction for Cybersickness Mitigation in Virtual Reality. *IEEE ISMAR 2025*.

Patents

[1] Cybersickness Reduction System for Head-Mounted Displays Using Eye-Tracking-Based Peripheral Vision Modulation. *KR Patent*, 2026.

Awards & Honors

2026 **Best Poster Honorable Mention Award [P1]**

IEEE VR'26, Korea

2026 **Best Ph.D. Graduation Award**

Kyung Hee University, Korea

2025 **Gold Prize, Research Performance Contest**

Kyung Hee University, Korea

2024 **Graduate Research Fellowship**

Kyung Hee University, Korea

Research Experience

[1] **Human-Centered Extended Reality Lab**, Kyung Hee University

Graduate Researcher

Mar. 2024 ~ Feb. 2026

Designed Unity-based VR experimental environments for cybersickness studies. Developed peripheral brightness and resolution modulation shaders using Gaussian spatial weighting. Integrated eye-tracking systems (Varjo XR-4) and physiological sensors (EDA). Conducted large-scale user studies (N=50) and performed statistical analyses including repeated-measures ANOVA, paired t-tests, and effect size estimation.

Selected Projects

[1] **Peripheral Brightness Dimming for VR-Based Digital Therapeutics**: Proposed a VR-based digital therapeutic framework for mitigating cybersickness while preserving presence through peripheral vision modulation.

Technical Skills

Programming: C#, Python, MATLAB

Engines & SDKs: Unity, SteamVR, Varjo SDK

Hardware: Varjo XR-4, HTC Vive Pro Eye

Analysis: Statistical Analysis, ANOVA, Effect Size, EDA Processing

Languages

Korean (Native), English (Intermediate)